Global Manufacturing Company



# Enterprise Data Governance Framework Development & Roadmap for a Global Manufacturing Company

Presented by **Felice Benita**

# Executive Summary (1)

## Business Context



The Global Manufacturing Company operates multiple manufacturing plants across different countries, supported by several enterprise platforms including ERP, MES, SCM, PLM, CRM, and IoT systems.
Rapid business growth, acquisitions, and decentralized operations have resulted in fragmented master data, inconsistent reporting, and limited enterprise-wide data visibility.

## Current Challenges:



- Multiple ERP instances with inconsistent master data
- Duplicate Material, Vendor, and Customer records
- Low confidence in executive reporting
- Manual reconciliation across plants
- No enterprise-wide Data Ownership
- Limited metadata and data lineage
- Inconsistent Data Quality Rules
- Increasing regulatory and audit requirements

# Executive Summary (2)

Enterprise data has become a strategic business asset, yet inconsistent governance limits its value.

The company currently manages data across multiple:

- ➡️ ERP systems
- ➡️ Manufacturing systems
- ➡️ CRM applications
- ➡️ Spreadsheets
- ➡️ Regional databases

The absence of a unified governance model has resulted in:

- Inconsistent master data
- Varying business definitions
- Duplicate records
- Limited data accountability

**Key Outcomes:**

- ✔ Standardized enterprise data
- ✔ Improved data quality
- ✔ Clear ownership
- ✔ Faster reporting
- ✔ Better compliance
- ✔ Trusted analytics

# Business Drivers

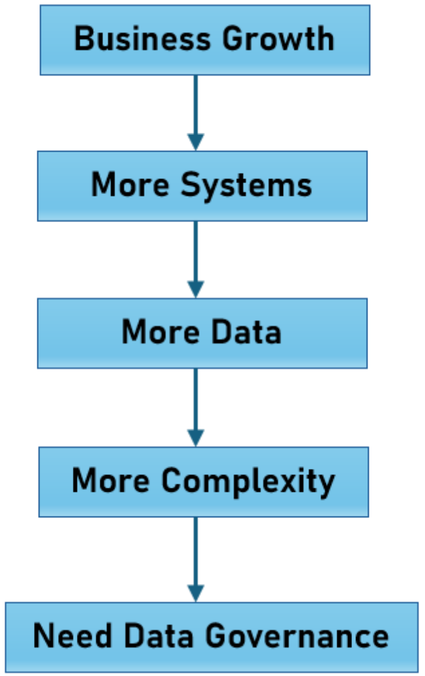
Rapid business growth requires trusted, standardized, and governed enterprise data.

Increasing business complexity has significantly expanded the volume, variety, and velocity of enterprise data.

**Key strategic initiatives include:**

- Global manufacturing operations
- Multi-country supply chain
- Customer-centric operations
- Digital transformation
- AI & Advanced Analytics
- ESG Reporting
- Regulatory Compliance
- Industry 4.0

Without strong governance, these initiatives face increased operational and compliance risks.




Business Drivers:

- 🏭 Manufacturing
- 🚚 Supply Chain
- 📦 Procurement
- 🌎 Global Expansion
- 📊 Analytics
- 🤖 AI

# Transformation Vision

- ☑ **Designed a comprehensive Enterprise Data Governance Framework** to establish clear governance structures, standardized policies, and defined data ownership across global manufacturing operations.
- ☑ **Strengthened enterprise data management capabilities** through Data Quality Management, Metadata Management, Master Data Management (MDM), Data Security, and Data Lifecycle Management to improve data accuracy, consistency, and compliance.
- ☑ **Developed an integrated enterprise data architecture** connecting ERP, MES, SCADA, Industrial IoT, PLM, SCM, WMS, and analytics platforms to enable a trusted **Single Source of Truth (SSOT)** and seamless end-to-end data flow.
- ☑ **Defined governance KPIs, success metrics, and a phased implementation roadmap** to measure governance maturity, monitor performance, and support continuous improvement across the organization.
- ☑ **Established a scalable, business-driven data foundation** that enhances operational excellence, supports digital manufacturing and Industry 4.0 initiatives, and enables trusted analytics, AI, and data-driven decision-making.

# Project Overview (1)

## Project Objectives

- ☑ Establish enterprise-wide data governance
- ☑ Define a clear **Target Operating Model (TOM)**
- ☑ Improve data quality
- ☑ Establish a **Single Source of Truth (SSOT)**
- ☑ Strengthen **data ownership, stewardship, and accountability**
- ☑ Enhance **data security, privacy, and regulatory compliance**
- ☑ Enable seamless integration across **ERP, MES, SCADA, Industrial IoT, PLM, SCM, WMS, and enterprise analytics platforms**
- ☑ Support **digital manufacturing & Industry 4.0**
- ☑ Enable AI/Analytics readiness



## Scope of Data Governance

### Business Functions

- Manufacturing Operations
- Supply Chain & Logistics
- Procurement
- Sales & Customer Management
- Finance & Controlling
- Engineering & Product Lifecycle Management (PLM)
- Quality Management
- Warehouse Management
- Maintenance & Asset Management
- Human Resources

# Project Overview (2)

## Scope of Data Governance

### Core Data Domains

- Material Master
- Product Master
- Bill of Materials (BOM)
- Routing & Production Master
- Supplier Master
- Customer Master
- Plant & Warehouse Master
- Equipment & Asset Master
- Employee Master
- Inventory Data
- Production Data
- Quality Inspection Data
- Maintenance Data
- Financial Data



## Scope of Data Governance

### Enterprise Systems

- Enterprise Resource Planning (ERP)
- Manufacturing Execution System (MES)
- Supervisory Control and Data Acquisition (SCADA)
- Industrial Internet of Things (IoT)
- Product Lifecycle Management (PLM)
- Supply Chain Management (SCM)
- Warehouse Management System (WMS)
- Customer Relationship Management (CRM)
- Enterprise Data Warehouse (EDW)
- Enterprise Data Lake
- Business Intelligence & Analytics Platforms

# Expected Business Outcomes

**FROM**
**Fragmented Data Landscape**

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**TO**
**Trusted Enterprise Data Platform**

**THROUGH**
**Standardized Governance**

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**High-Quality Data**

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**Global Visibility**

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**AI & Advanced Analytics Ready**

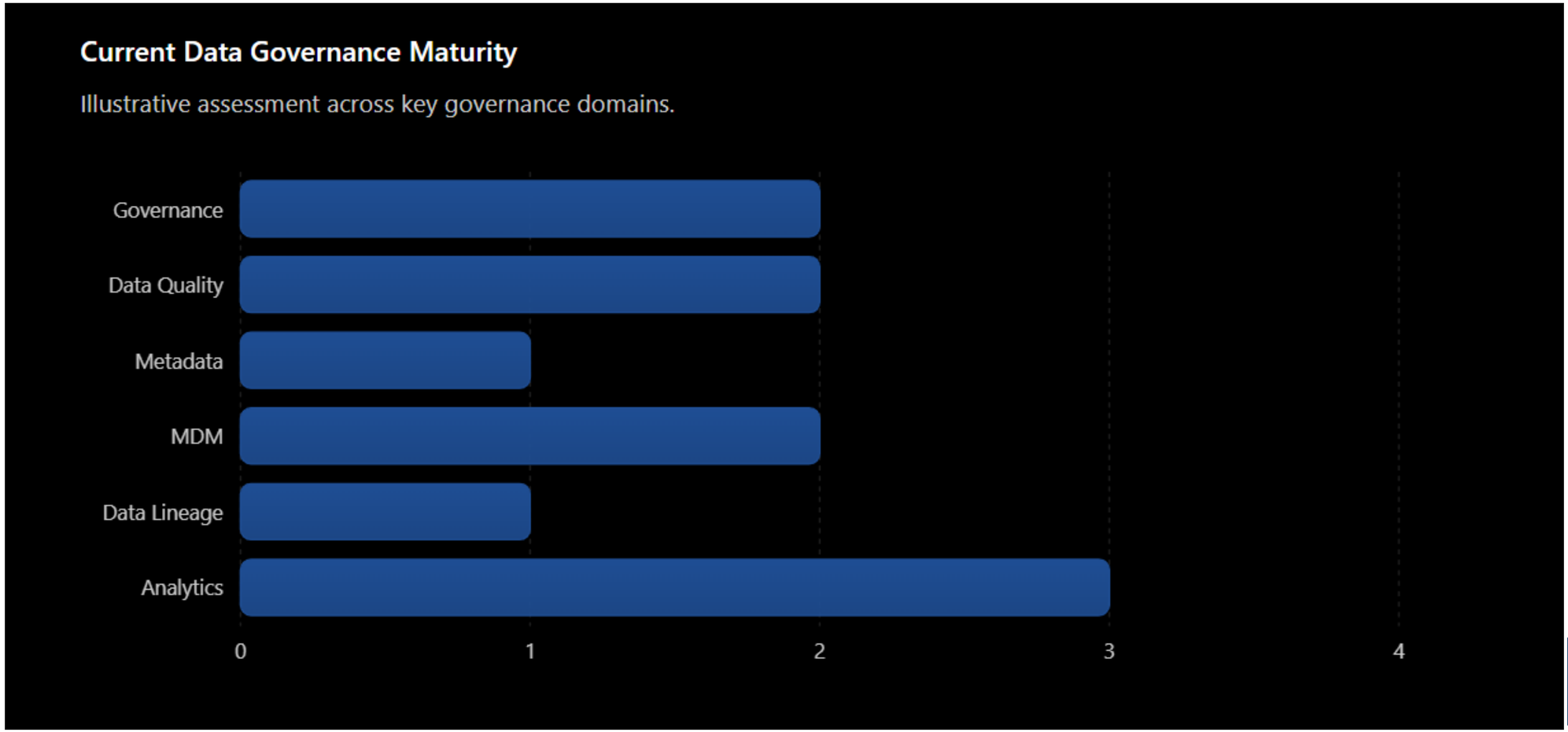
| Strategic Objective | Expected Benefit |
|---|---|
| Improve Data Quality | >95% Enterprise Data Quality Score |
| Increase Accountability | 100% Data Ownership Coverage |
| Reduce Master Data Duplication | >30% Reduction |
| Improve Reporting | Single Source of Truth |
| Accelerate Decision-Making | Faster Executive Reporting |
| Strengthen Compliance | Improved Audit Readiness |



# Current State Assessment (AS-IS)

| Business Area | Current Challenges | Business Impact |
|---|---|---|
| Master Data | Duplicate Materials | Inventory Inaccuracy |
| Manufacturing | Inconsistent BOM | Production Delays |
| Supply Chain | Duplicate Vendors | Procurement Risk |
| Sales | Duplicate Customers | Poor Customer Experience |
| Finance | Different Reporting Logic | Inconsistent Financial Reports |
| Analytics | Multiple Data Definitions | Low Executive Confidence |

# Current Data Governance Maturity




Target: 5

# Gap Assessment (1)

## Gap Analysis Matrix

| Capability | Current State | Target State | Priority |
|---|---|---|---|
| Data Governance Organization | Department-based | Enterprise-wide Governance Office | High |
| Data Ownership | Partial | 100% Assigned | High |
| Data Stewardship | Informal | Formal Steward Network | High |
| Data Quality | Manual | Automated Monitoring | High |
| Metadata Management | Spreadsheet-based | Enterprise Data Catalog | High |
| Master Data Management | Multiple Versions | Golden Record | High |
| Data Lineage | Limited | End-to-End Lineage | Medium |
| Governance KPIs | Not Defined | Executive Dashboard | Medium |
| Data Architecture | Fragmented | Integrated Enterprise Architecture | High |

# Gap Assessment (2)

## Capability Gap Analysis

| Domain | Current | Target | Gap |
|---|---|---|---|
| Governance | 2/5 | 5/5 | High |
| Data Quality | 2/5 | 5/5 | High |
| Metadata | 1/5 | 5/5 | Very High |
| MDM | 2/5 | 5/5 | High |
| Data Lineage | 1/5 | 5/5 | Very High |
| Analytics | 3/5 | 5/5 | Medium |



# Root Causes

Lack of Enterprise Governance

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No Standard Policies

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Inconsistent Business Rules

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Poor Data Quality

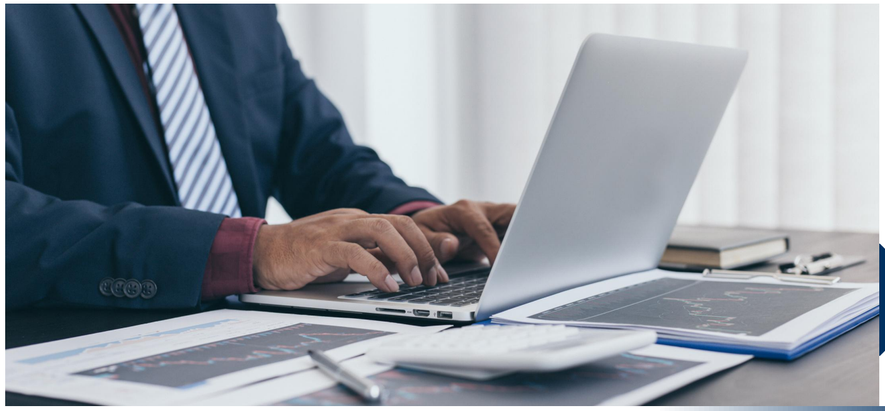
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Low Reporting Confidence

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Operational Inefficiency





# Target Operating Model (Federated)



**Governance Model Type:**

## Federated Governance Model

Recommended for a Global Manufacturing Company because:

- Global standardization of data policies and definitions
- Local accountability through Data Owners and Data Stewards
- Faster issue resolution within business domains
- Consistent master data across plants and regions
- Better support for mergers, acquisitions, and new manufacturing sites
- Improved data quality and trusted reporting
- Scalable governance as the organization grows

# Target Operating Model (Federated)

Link to: 📜 DATA GOVERNANCE TARGET OPERATING MODEL (TOM)



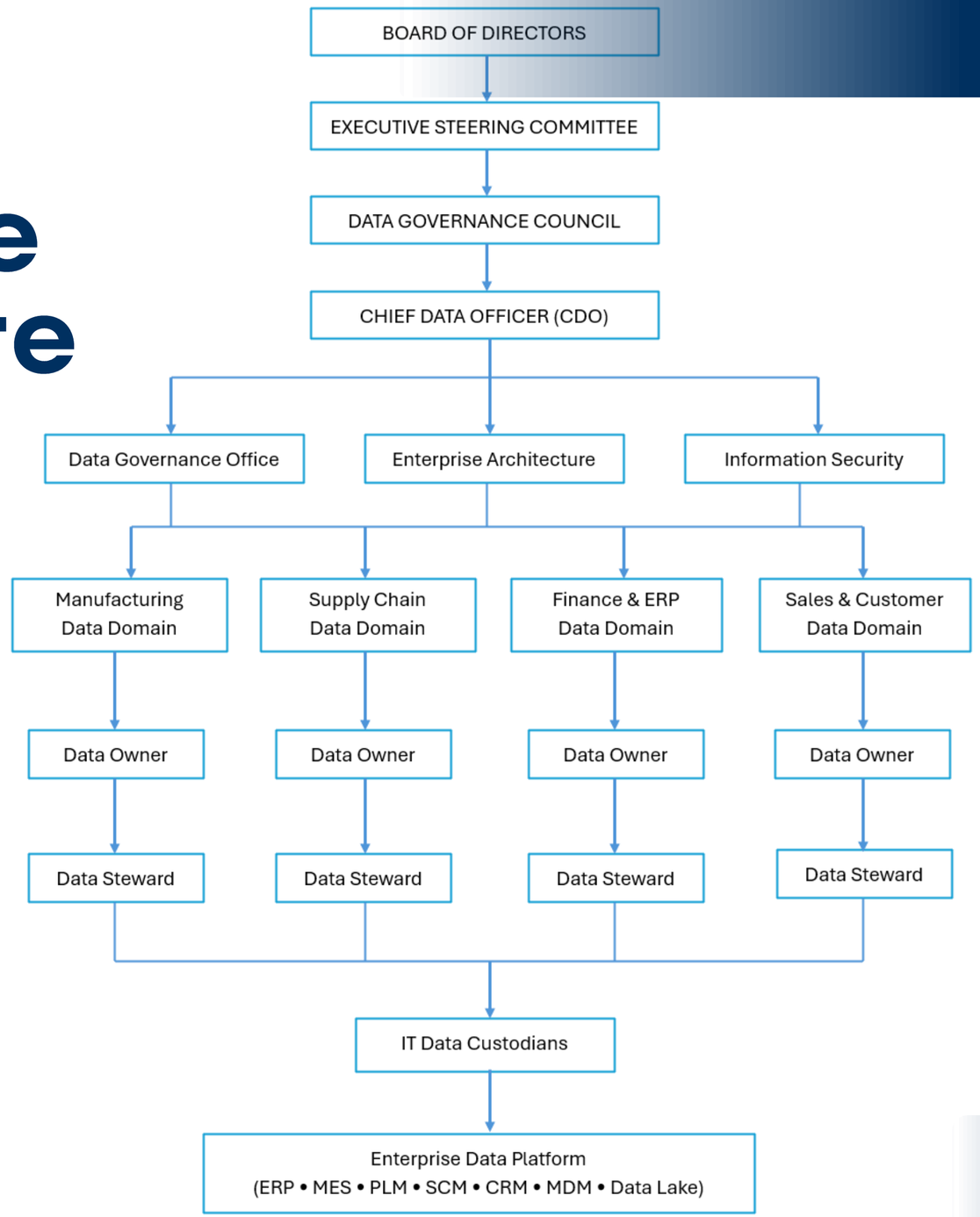


# RACI Matrix

| Governance Activity | DGC | CDO | DGO | DO | DS | IT | EA | SEC |
|---|---|---|---|---|---|---|---|---|
| Approve Data Governance Policy | A | R | C | I | I | I | I | C |
| Approve Enterprise Standards | A | R | C | C | I | I | C | C |
| Define Business Glossary | I | C | R | A | R | I | C | I |
| Define Critical Data Elements | I | C | R | A | R | I | I | I |
| Data Quality Rules | I | C | C | A | R | C | I | I |
| Master Data Governance | I | C | R | A | R | C | C | I |
| Metadata Management | I | C | A | R | R | C | C | I |
| Data Quality Monitoring | I | C | C | A | R | C | I | I |
| Issue Resolution | I | C | C | A | R | C | I | I |
| KPI Reporting | C | A | R | C | C | I | I | I |
| Access Approval | I | C | I | A | C | R | I | C |
| Security Controls | I | I | I | C | I | R | C | A |

**Legend:** R = Responsible, A = Accountable, C = Consulted, I = Informed

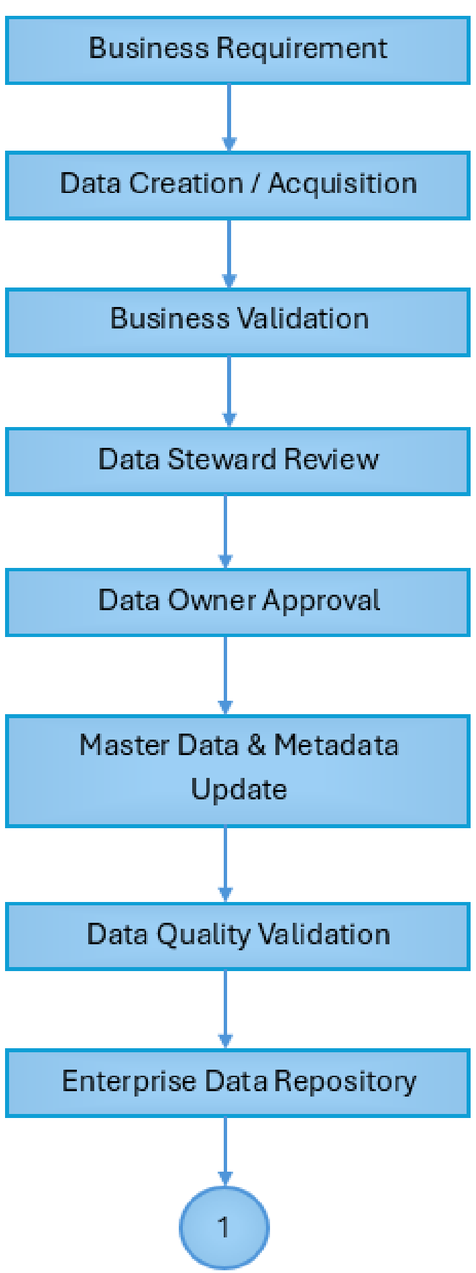
# Data Governance Architecture

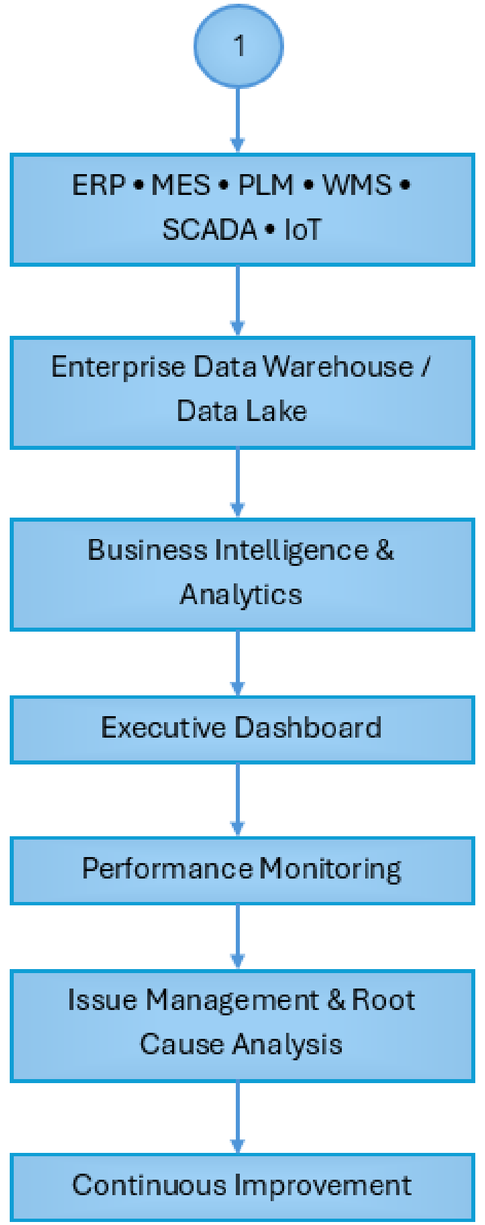



Zoom Diagram

# Governance Flow Diagram

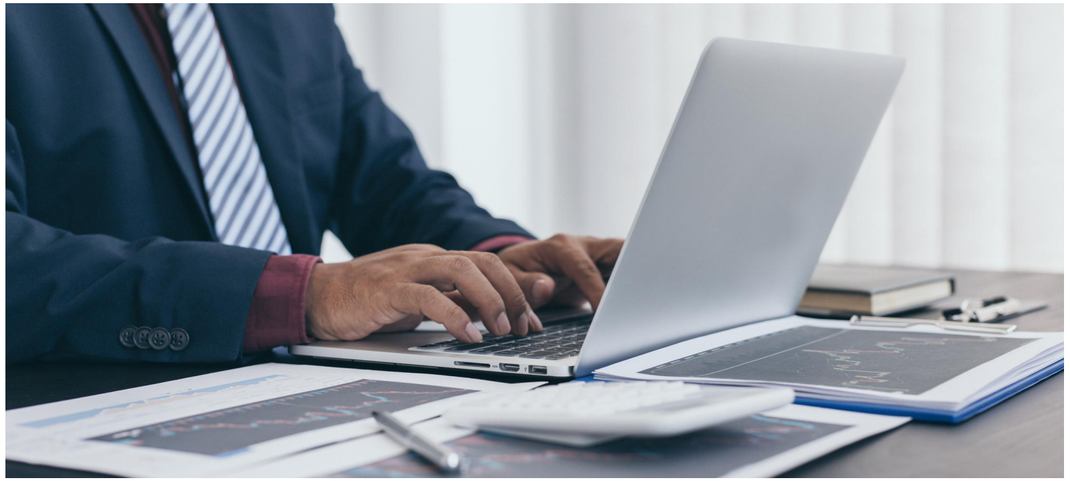
Zoom Diagram



# Data Governance Policy

Link to: 📜 DATA GOVERNANCE POLICY



# Data Quality Management Framework

**Vision:**

Deliver trusted, accurate, complete, and timely enterprise data to support operational excellence, global manufacturing execution, supply chain optimization, regulatory compliance, and data-driven decision-making.

**Framework Overview**



# Data Quality Operating Model

Business Process

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Data Creation

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Validation Rules

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Data Quality Engine

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Monitoring

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Issue Detection

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Steward Review

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Remediation

↓

Continuous Improvement

## Governance Roles



| Role | Responsibility |
|---|---|
| Data Owner | Accountable for data quality |
| Data Steward | Daily monitoring and remediation |
| IT Data Custodian | Technical controls and automation |
| Data Governance Office | Governance oversight |
| Data Governance Council | KPI review and strategic decisions |

# Data Quality Dimensions

**Enterprise Standards**

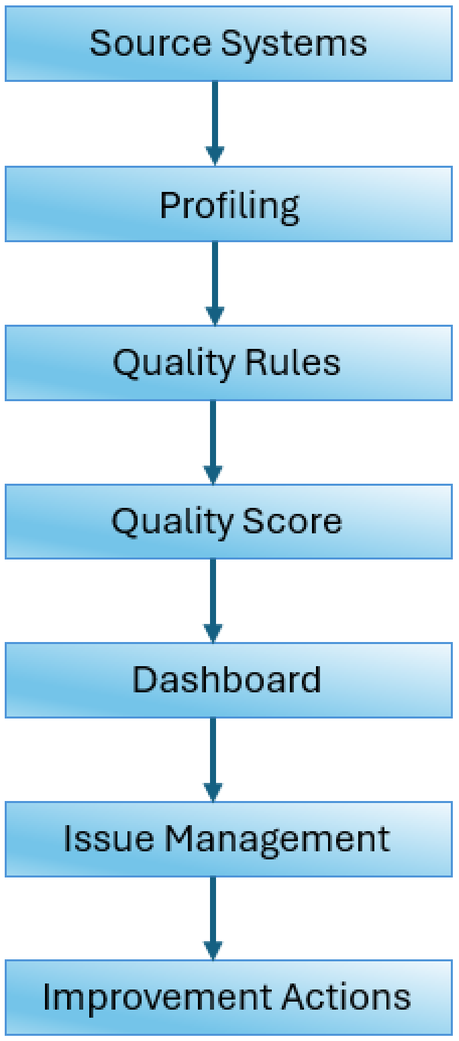
| Dimension | Manufacturing Example | Target |
|---|---|---|
| Accuracy | Correct material specifications | ≥98% |
| Completeness | Mandatory BOM fields populated | ≥95% |
| Consistency | Same product code across ERP and MES | ≥98% |
| Timeliness | Production data updated within SLA | ≥95% |
| Validity | Values comply with business rules | ≥99% |
| Uniqueness | No duplicate supplier records | ≥99.5% |

# Data Quality Rule Library

| Data Domain | Example Rule |
|---|---|
| Material | Material ID must be unique |
| Product | Product status must reference approved values |
| BOM | Components must exist in Material Master |
| Supplier | Supplier tax ID must be unique |
| Inventory | Stock quantity cannot be negative |
| Purchase Order | Supplier must exist before PO creation |
| Equipment | Asset ID must be unique |
| Production Order | Plant code must be valid |



# Data Quality Monitoring






# Implementation Roadmap - Overview

**Executive Roadmap Overview**

| Phase | Timeline | Primary Objective |
|---|---|---|
| **Phase 1** | Months 1–3 | Governance Foundation |
| **Phase 2** | Months 4–6 | Governance Standardization |
| **Phase 3** | Months 7–12 | Technology Enablement & Enterprise Rollout |
| **Phase 4** | Months 13–18 | Optimization & Continuous Improvement |

# Implementation Roadmap (1)

## ☑ Phase 1

### Governance Foundation (Months 1–3)

Deliverables:

- Governance Charter
- Governance Organization
- Operating Model
- Policy Framework
- Governance Roles
- Current State Report
- Gap Assessment
- Data Domain Catalogue



## ☑ Phase 2

### Governance Standardization (Months 4–6)

Deliverables:

- Enterprise Business Glossary
- Metadata Repository
- Data Catalog
- Data Quality Rules
- MDM Standards
- Governance Procedures
- Standard Operating Procedures (SOPs)

# Implementation Roadmap (2)

## ☑ Phase 3

### Technology Enablement & Enterprise Rollout (Months 7–12)

Deliverables:

- Production-ready Governance Platform
- Data Catalog
- MDM Hub
- Automated Data Quality Monitoring
- KPI Dashboard
- Governance Training Materials
- Change Management Plan

## ☑ Phase 4

### Optimization & Continuous Improvement (Months 13–18)

Deliverables:

- Governance Performance Dashboard
- Continuous Improvement Register
- Updated Governance Policies
- AI-enabled Data Quality Capabilities
- Governance Maturity Assessment Report
- Executive Performance Report

# Enterprise Data Architecture

**GLOBAL MANUFACTURING ENTERPRISE**

**Business Applications**

ERP MES PLM SCM CRM HCM
WMS QMS EAM TMS SRM

**Operational Technology (OT)**

SCADA
PLC
Industrial IoT
Production Machines
Robotics
Sensors

**Enterprise Integration Layer**

API Gateway
Enterprise Service Bus (ESB)
Event Streaming
ETL / ELT
Data Integration Platform

**Enterprise Data Platform**

Enterprise Data Lake
Enterprise Data Warehouse
Master Data Management
Metadata Repository
Business Glossary
Data Catalog
Data Quality Engine

**Analytics & Consumption**

Power BI
Tableau
Executive Dashboard
Self-Service BI
Machine Learning
AI
Predictive Analytics

## Architecture Principles:

- One enterprise data platform
- Centralized governance with federated ownership
- Single Source of Truth (SSOT)
- Event-driven integration where appropriate
- Standardized APIs and data contracts
- Metadata-driven architecture
- Security by design
- Scalable cloud-ready foundation

# KPIs & Success Metrics (1)

| KPI | Before (Current State) | Target (Future State) | Success Criteria |
|---|---|---|---|
| Enterprise Data Quality Score | ~78% | **≥95%** | Enterprise data consistently meets quality thresholds across all critical data domains. |
| Critical Data Elements (CDEs) with Assigned Data Owner | 25% | **100%** | Every Critical Data Element has a formally assigned Data Owner accountable for governance. |
| Business Domains with Assigned Data Steward | 30% | **100%** | All business domains have designated Data Stewards actively performing governance activities. |
| Metadata Coverage | 20% | **≥95%** | Critical enterprise data assets are documented in the enterprise metadata repository and data catalog. |
| Business Glossary Completion | 15% | **100%** | Enterprise business terms are standardized, approved, and published across all business functions. |
| Master Data Golden Record Coverage | 45% | **≥98%** | Master data entities (Material, Supplier, Customer, Product, Plant) are managed as Golden Records. |
| Duplicate Material Records | 8% | **<0.5%** | Duplicate material records are minimized through MDM governance and automated matching. |
| Duplicate Supplier Records | 6% | **<0.5%** | Supplier master data duplication is effectively eliminated. |
| Duplicate Customer Records | 7% | **<1%** | Customer master data maintains high uniqueness and consistency. |
| Data Quality Rule Coverage | 35% | **100%** | All Critical Data Elements have documented and implemented Data Quality Rules. |
| Automated Data Quality Monitoring | 20% | **100%** | Enterprise Data Quality monitoring is automated for all critical data domains. |
| Data Issue SLA Compliance | 65% | **≥95%** | Data quality issues are resolved within agreed service-level targets. |

# KPIs & Success Metrics (2)

| KPI | Before (Current State) | Target (Future State) | Success Criteria |
|---|---|---|---|
| Critical Data Issues Open >30 Days | 18 | **0** | No unresolved critical data issues exceed 30 calendar days. |
| Policy Compliance Rate | 60% | **≥98%** | Business units consistently comply with approved Data Governance Policies. |
| Quarterly Data Steward Review Completion | 40% | **100%** | All scheduled stewardship reviews are completed on time. |
| Quarterly Data Owner Certification | Not Performed | **100%** | Data Owners formally certify data ownership, quality, and compliance each quarter. |
| Enterprise Data Catalog Adoption | 10% | **≥90%** | Business users actively use the enterprise data catalog to discover and understand data assets. |
| Data Lineage Coverage | 5% | **≥90%** | End-to-end lineage is documented for critical reports and data assets. |
| Sensitive Data Classification Coverage | 40% | **100%** | All sensitive and confidential data assets are classified according to enterprise policy. |
| Access Review Completion | 55% | **100%** | Quarterly access reviews are completed for all governed systems and sensitive data. |
| High Severity Data Security Incidents | 5/year | **0** | No critical data security or privacy incidents attributable to governance failures. |
| Audit Findings Related to Data Governance | 12 Major Findings | **0 Major Findings** | Internal and external audits report no major governance deficiencies. |
| ERP–MES Master Data Synchronization Success Rate | 82% | **≥99%** | Master data synchronization between enterprise systems is reliable and consistent. |
| Manufacturing Master Data Accuracy | 88% | **≥99%** | Material, BOM, Routing, and Plant master data accurately reflect approved business records. |

# KPIs & Success Metrics (3)

| KPI | Before (Current State) | Target (Future State) | Success Criteria |
|---|---|---|---|
| Executive Report Consistency | 70% | **100%** | Executive reports are generated from a single governed source with consistent definitions. |
| Time to Identify Data Issues | 5 days | **<4 hours** | Critical data issues are detected rapidly through automated monitoring. |
| Time to Resolve Critical Data Issues | 15 days | **<24 hours** | Critical data issues are resolved within established governance SLAs. |
| Self-Service Analytics Adoption | 20% | **≥80%** | Business users confidently access governed data for analytics without IT intervention. |
| Business User Trust in Enterprise Data | 60% | **≥95%** | Survey results indicate high confidence in enterprise data quality and reliability. |
| Enterprise Governance Maturity (DAMA/DCAM-aligned) | Level 2 (Developing) | **Level 4 (Managed)** | Governance capabilities are standardized, measured, and continuously improved across the enterprise. |

# Conclusion & Recommendations



The proposed **Enterprise Data Governance Framework** provides a comprehensive and scalable approach to managing data as a strategic enterprise asset across the Global Manufacturing Company. By establishing clear governance structures, standardized policies, defined data ownership, and integrated data management capabilities, the organization can improve data quality, operational efficiency, regulatory compliance, and decision-making. Combined with a modern enterprise data architecture integrating **ERP, MES, SCADA, and IoT**, the framework creates a trusted data foundation that supports digital transformation, Industry 4.0 initiatives, and future AI-driven manufacturing capabilities.



Global Manufacturing Company

# THANK YOU

Presented by **Felice Benita**